

Expressions, Formulae and Brackets — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to come out exactly.
- The rules for expanding, factorising and signs are not printed on this paper. You are expected to know them.
- When a bracket is subtracted, the minus sign multiplies every term inside it. This is the single largest source of lost marks in this topic.
- Several questions ask you to explain or to prove something is always true. A calculation on its own is not an explanation — say what the algebra shows.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – BRACKETS THAT ARE TAKEN AWAY

1. Expand and simplify:

(a) $5(2x + 3) - 2(x + 4)$ [2]

(b) $4(3x - 2) - 3(2x - 5)$ [2]

2. Expand and simplify:

(a) $3(2x - 5) - (x - 4)$ [2]

(b) $2(4x + 1) - 5(x - 3)$

[2]

3. Given that $a = -3$ and $b = 4$, work out:

(a) a^2

[1]

(b) $2a^2$

[1]

(c) $(a + b)^2$

[1]

(d) $a^2 + b^2$

[1]

SECTION 2 – FACTORISING COMPLETELY

4. Factorise completely. One of these cannot be factorised — say so and explain why.

(a) $18x + 24$

[1]

(b) $15y^2 - 20y$

[1]

(c) $12a^2 + 18a$

[1]

(d) $8x + 3y$

[1]

- 5.** A rectangular garden has an area of $(20x + 45) \text{ m}^2$. One of its sides is 5 m long.

(a) Factorise $20x + 45$ completely.

[2]

(b) Hence write down an expression for the other side.

[1]

(c) Write an expression for the perimeter, fully simplified.

[1]

SECTION 3 – SHOWING SOMETHING IS ALWAYS TRUE

- 6.** Two consecutive whole numbers can be written as n and $n + 1$.

(a) Write an expression for their sum, fully simplified.

[1]

(b) Use your answer to explain why the sum of two consecutive whole numbers is always odd.

[2]

- (c)** Write a simplified expression for the sum of three consecutive whole numbers, [2]
and use it to explain why that sum is always a multiple of 3.

- 7.** The cost C rupees of hiring a hall for h hours is given by the formula $C = 150 + 25h$.

- (a)** Find the cost of hiring the hall for 6 hours. [1]

- (b)** A booking cost 425 rupees. For how many hours was the hall hired? [2]

- (c)** Explain why a booking can never cost exactly 210 rupees if the hall is only hired for a whole number of hours. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.